HDOC - 72nd day

DYNAMICALLY RESIZABLE STUDENT ATTENDANCE MATRIX

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Problem Statement

Develop a C program to manage student attendance using a dynamically resizable matrix. Each row represents a student and each column represents a working day. Use 1 for Present and 0 for Absent. Dynamically allocate the initial matrix using malloc(), allow new students to be added using realloc(), calculate each student's attendance percentage, use separate functions for the required operations, validate attendance input, and release all dynamically allocated memory using free().

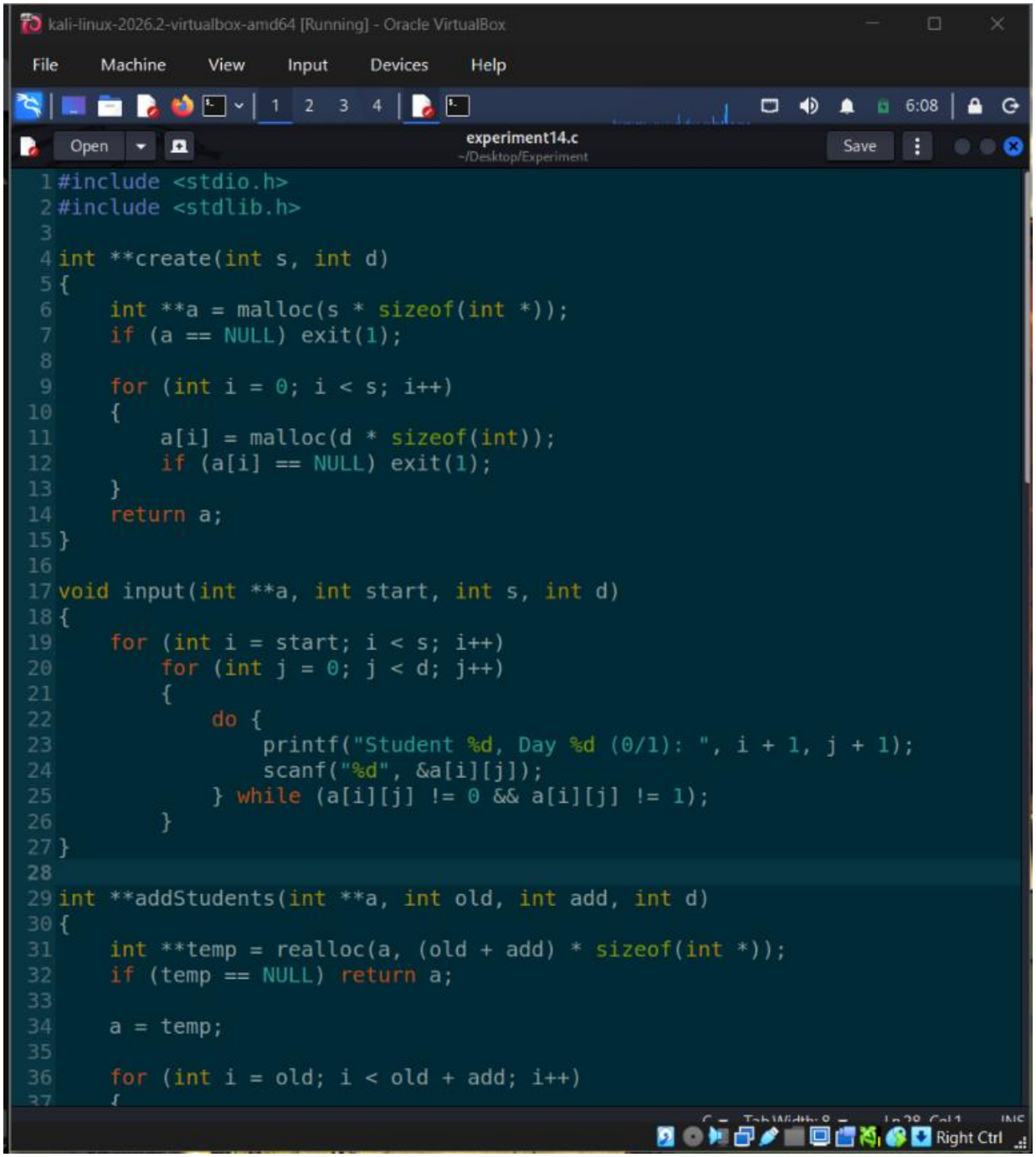
Algorithm

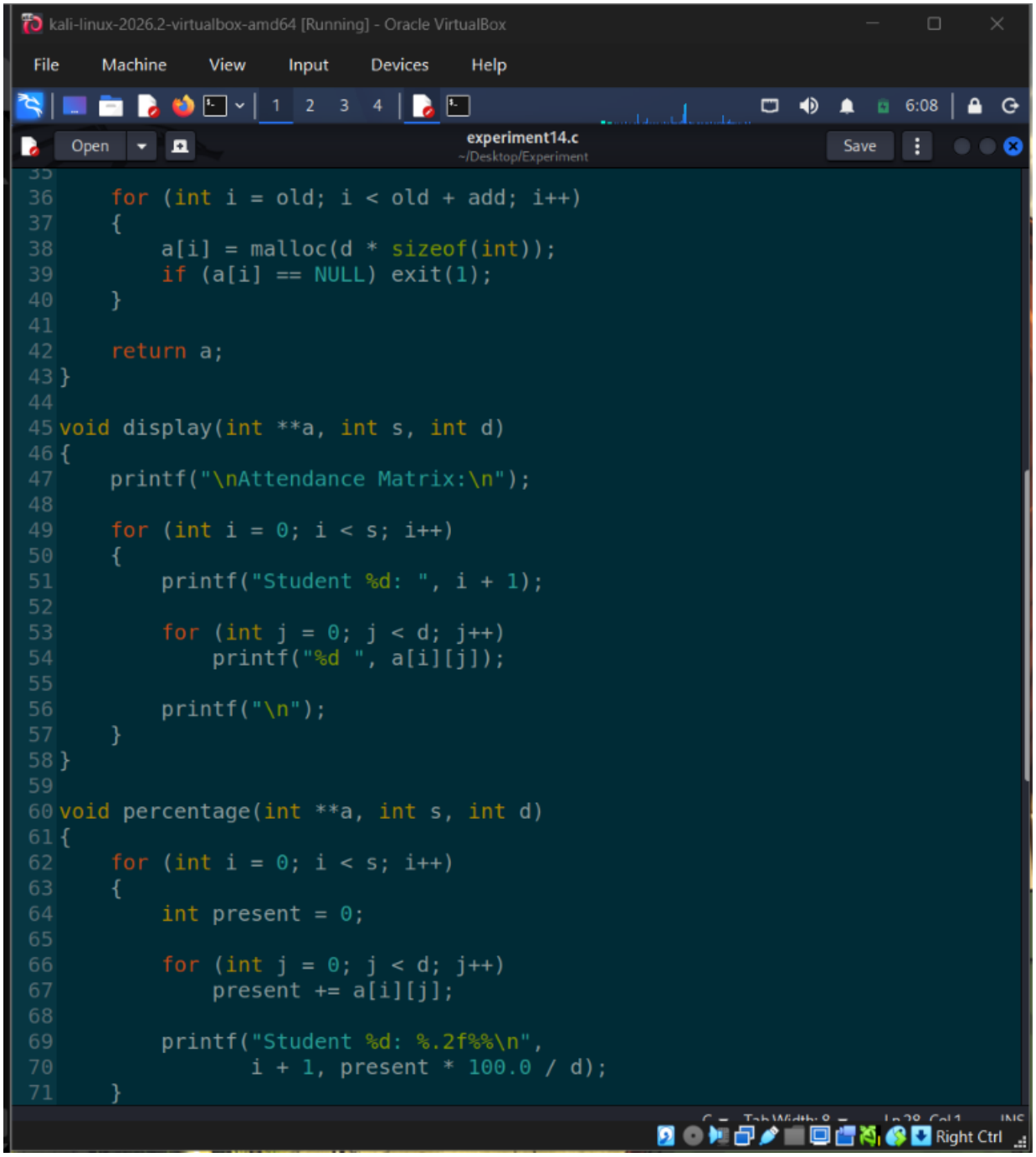
1. Start.
2. Read the initial number of students and working days.
3. Dynamically create the attendance matrix using malloc().
4. Record attendance for each student and accept only 0 or 1.
5. Read the number of additional students to be added.
6. Use realloc() to increase the matrix and allocate rows for the new students.
7. Record attendance for the newly added students.
8. Display the complete attendance matrix.
9. Calculate and display the attendance percentage of each student.
10. Release all dynamically allocated memory using free() and stop.

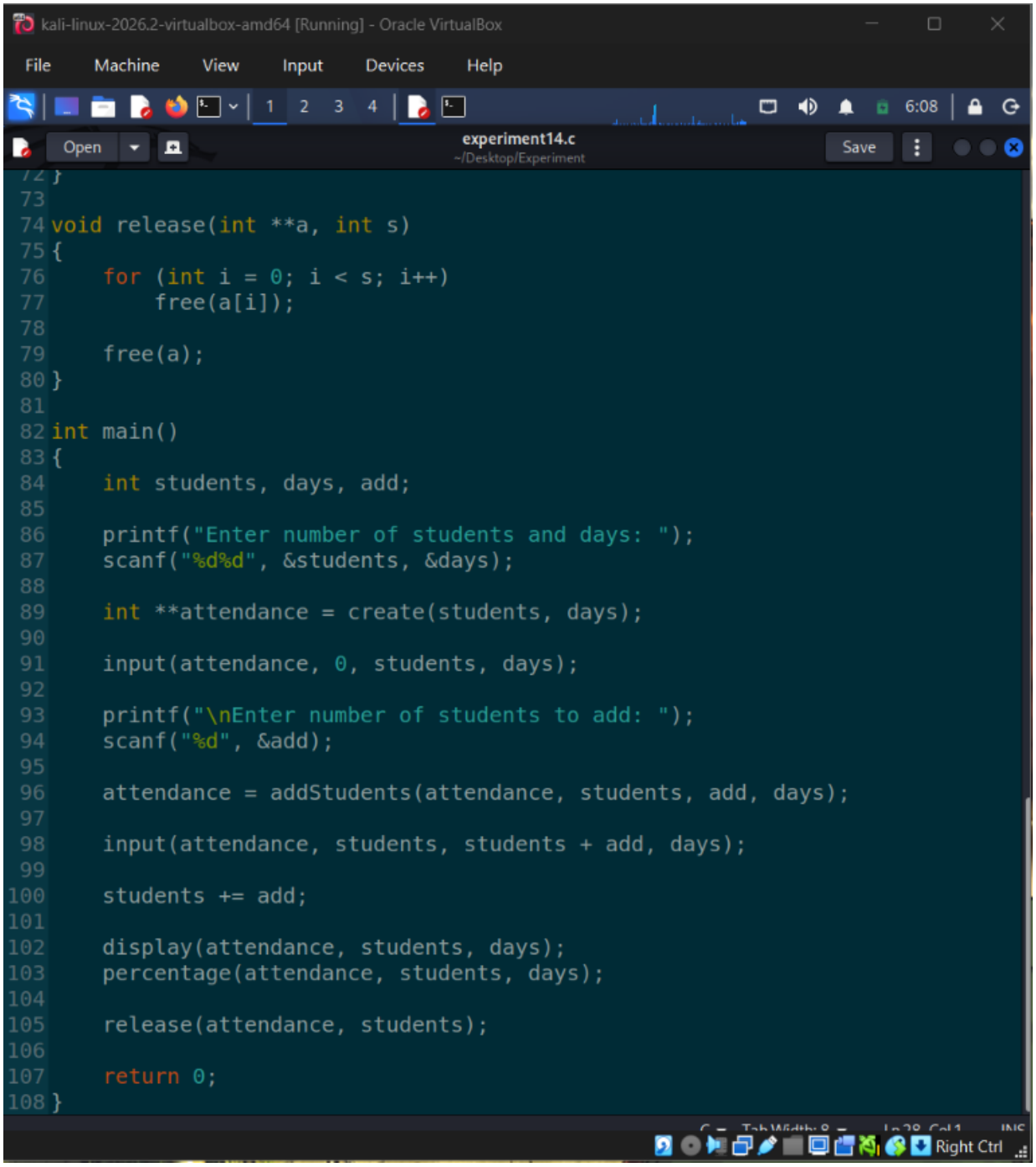
Test Case Table

S. No.	Input	Expected Output	Actual Output	Result
1	Students = 2, Days = 5 S1: 1 0 1 1 0 S2: 1 1 0 1 1 Add = 0	Matrix: S1: 1 0 1 1 0 S2: 1 1 0 1 1 Percentages: 60.00%, 80.00%	Matrix: S1: 1 0 1 1 0 S2: 1 1 0 1 1 Percentages: 60.00%, 80.00%	Pass
2	Students = 2, Days = 6 S1: 1 1 1 0 0 1 S2: 0 1 0 1 1 1 Add = 1 S3: 1 0 0 0 1 1	Matrix includes 3 students S1: 66.67% S2: 66.67% S3: 50.00%	Matrix includes 3 students S1: 66.67% S2: 66.67% S3: 50.00%	Pass

C Program


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Open
experiment14.c
~/Desktop/Experiment
Save
#include <stdio.h>
#include <stdlib.h>
int **create(int s, int d)
{
int **a = malloc(s * sizeof(int *));
if (a == NULL) exit(1);
for (int i = 0; i < s; i++)
{
a[i] = malloc(d * sizeof(int));
if (a[i] == NULL) exit(1);
}
return a;
}
void input(int **a, int start, int s, int d)
{
for (int i = start; i < s; i++)
for (int j = 0; j < d; j++)
{
do {
printf("Student %d, Day %d (0/1): ", i + 1, j + 1);
scanf("%d", &a[i][j]);
} while (a[i][j] != 0 && a[i][j] != 1);
}
}
int **addStudents(int **a, int old, int add, int d)
{
int **temp = realloc(a, (old + add) * sizeof(int *));
if (temp == NULL) return a;
a = temp;
for (int i = old; i < old + add; i++)
Right Ctrl


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
6:08
Open
experiment14.c
~/Desktop/Experiment
Save
for (int i = old; i < old + add; i++)
{
a[i] = malloc(d * sizeof(int));
if (a[i] == NULL) exit(1);
}
return a;
}
void display(int **a, int s, int d)
{
printf("\nAttendance Matrix:\n");
for (int i = 0; i < s; i++)
{
printf("Student %d: ", i + 1);
for (int j = 0; j < d; j++)
printf("%d ", a[i][j]);
printf("\n");
}
}
void percentage(int **a, int s, int d)
{
for (int i = 0; i < s; i++)
{
int present = 0;
for (int j = 0; j < d; j++)
present += a[i][j];
printf("Student %d: %.2f%%\n",
i + 1, present * 100.0 / d);
}
Right Ctrl

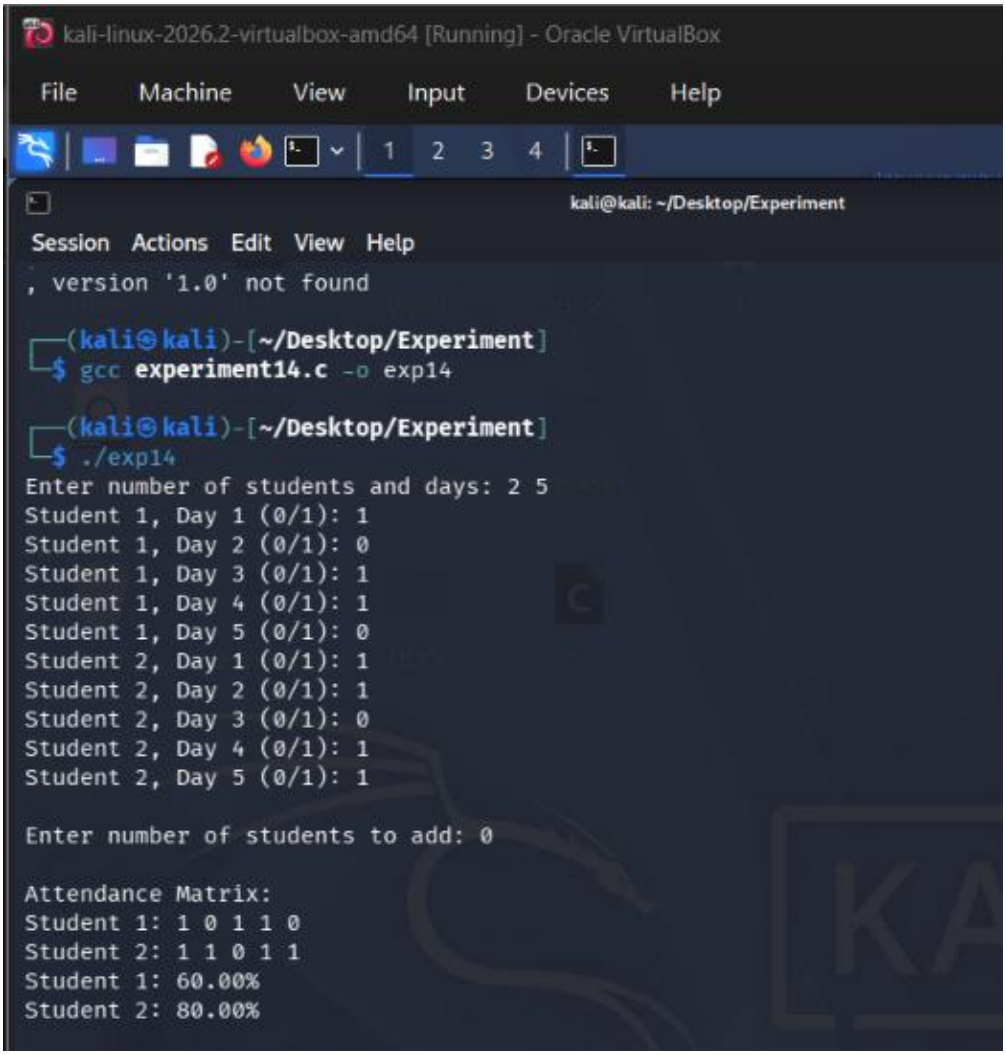

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
6:08
Open
experiment14.c
~/Desktop/Experiment
Save
}
void release(int **a, int s)
{
for (int i = 0; i < s; i++)
free(a[i]);
free(a);
}
int main()
{
int students, days, add;
printf("Enter number of students and days: ");
scanf("%d%d", &students, &days);
int **attendance = create(students, days);
input(attendance, 0, students, days);
printf("\nEnter number of students to add: ");
scanf("%d", &add);
attendance = addStudents(attendance, students, add, days);
input(attendance, students, students + add, days);
students += add;
display(attendance, students, days);
percentage(attendance, students, days);
release(attendance, students);
return 0;
}
Right Ctrl

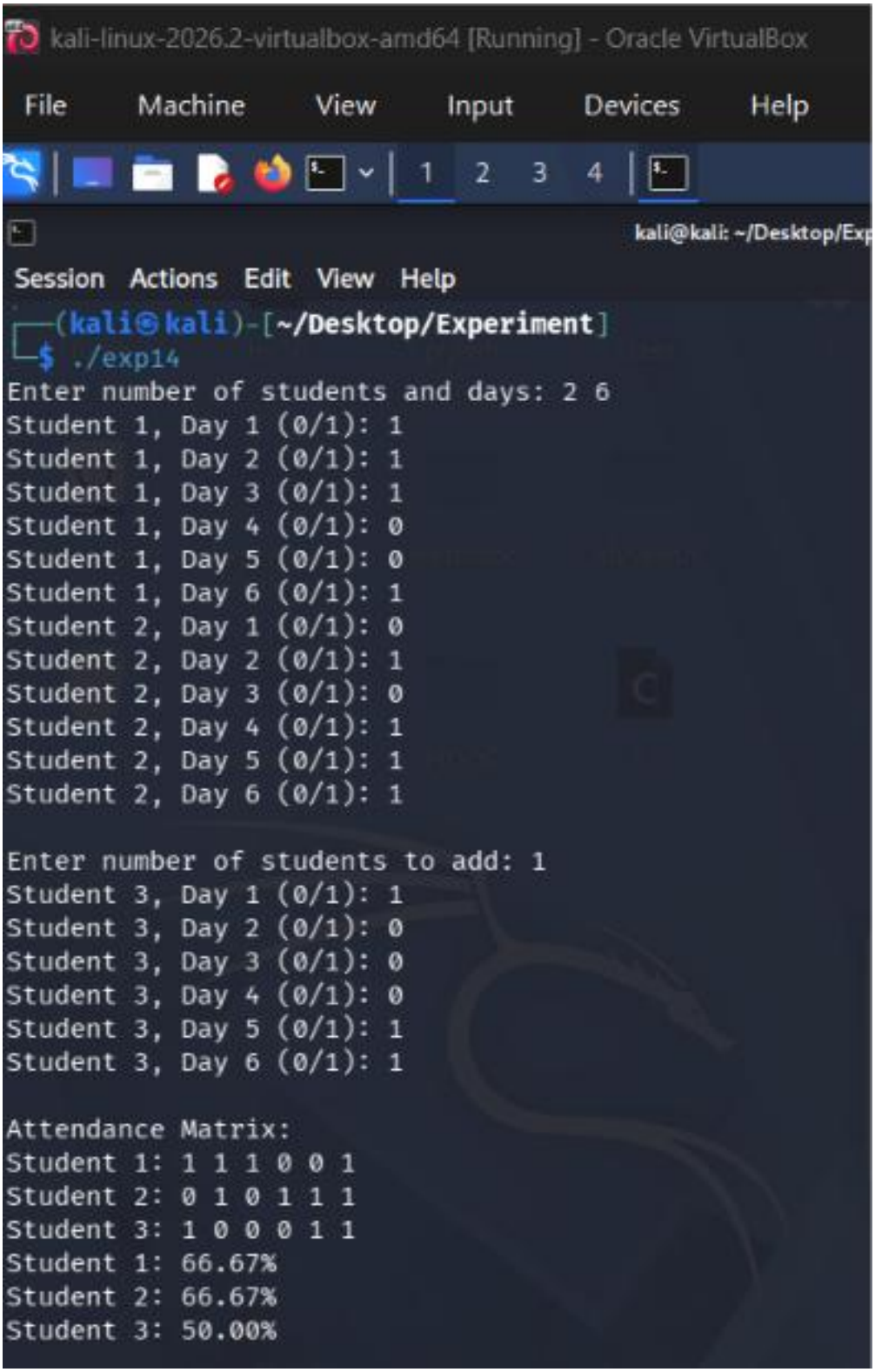
Compilation and Execution

Compilation command: gcc experiment14.c -o exp14

Execution command: ./exp14

Execution Screenshots:




kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali: ~/Desktop/Exp
Session Actions Edit View Help
(kali㉿kali)-[~/Desktop/Experiment]
$ ./exp14
Enter number of students and days: 2 6
Student 1, Day 1 (0/1): 1
Student 1, Day 2 (0/1): 1
Student 1, Day 3 (0/1): 1
Student 1, Day 4 (0/1): 0
Student 1, Day 5 (0/1): 0
Student 1, Day 6 (0/1): 1
Student 2, Day 1 (0/1): 0
Student 2, Day 2 (0/1): 1
Student 2, Day 3 (0/1): 0
Student 2, Day 4 (0/1): 1
Student 2, Day 5 (0/1): 1
Student 2, Day 6 (0/1): 1
Enter number of students to add: 1
Student 3, Day 1 (0/1): 1
Student 3, Day 2 (0/1): 0
Student 3, Day 3 (0/1): 0
Student 3, Day 4 (0/1): 0
Student 3, Day 5 (0/1): 1
Student 3, Day 6 (0/1): 1
Attendance Matrix:
Student 1: 1 1 1 0 0 1
Student 2: 0 1 0 1 1 1
Student 3: 1 0 0 0 1 1
Student 1: 66.67%
Student 2: 66.67%
Student 3: 50.00%